

IDBI BANK

Innovate 2026

Project

DISHA

Prospect Assist AI

PS-2: AI-Powered Lead Scoring & RM Copilot

Team Name

NeuralOps

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Problem Statement

PS-2: Prospect Assist AI

Track

Banking Innovation — Retail Lending

Submission Date

July 2026

Brief About the Idea

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AI-powered lead intelligence platform for IDBI Bank Relationship Managers

- No data-driven lead scoring — RMs rely on gut

The Problem

- Cold prospects go stale without re-engagement
- Zero institutional memory per prospect per RM
- Manual product matching erodes client trust
- No audit trail — compliance gaps in decisions

- ✓ Scores 386+ prospects on 0–100 scale (7-component)

The Solution — Project Disha

- ✓ Recommends best product × channel × timing (NBA)
- ✓ Answers RM questions via AI copilot (natural language)
- ✓ Re-engages cold prospects with personalised scripts
- ✓ Append-only audit logs with DB-level immutability

"Right prospect. Right product. Right moment."

Opportunities & USP

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How Disha stands apart from existing solutions

Market Opportunity

- IDBI retail pipeline has high lead attrition
- ■ billions in untapped cross-sell per RM
- Human capacity to score 500+ prospects → 0

Differentiation

- Explainable Scores — RM sees WHY (SHAP-style)
- Multi-agent: scoring + NBA + copilot parallel
- Org-specific weights trained on IDBI patterns

USP

- DVR loop: draft → verify → refine every answer
- Episodic memory per prospect across RM sessions
- Cold re-engagement via WhatsApp / SMS scripts

How Disha Solves the Problem

1. Ingest

Pluggable adapter reads prospect data

2. Score

7-component → 0–100 band
(Hot/Warm/Lukewarm/Cold)

3. Recommend

NBA: product × channel × timing per prospect

4. Explain

SHAP-style breakdown shows score drivers

5. Converse

DVR-loop copilot with verified, grounded replies

6. Re-engage

Urgency agent: WhatsApp + SMS for cold prospects

Key Features

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Five production-grade AI capabilities built into Disha

- 7-component weighted scorecard (income, CIBIL, digital activity, recency, segment, enquiries, life event)
- Deterministic + explainable — no black-box models
- SHAP-style signed contribution breakdown

- ProspectOrchestrator → ScoringAgent + NBAAgent
- Parallel execution via `asyncio.gather()`
- Next Best Action: product × channel × timing
- Priority reason from top score driver

- System-2: Draft → Verify → Refine every query
- Claude tool-use (up to 4 rounds) for data lookup
- Second LLM pass verifies factual accuracy
- Multi-provider: Claude → Gemini → rule-based

- CopilotMemory persists RM↔Disha turns
- Last 6 turns injected into every copilot call
- Memory is RM-scoped — isolated per RM
- Full history retrieval via API

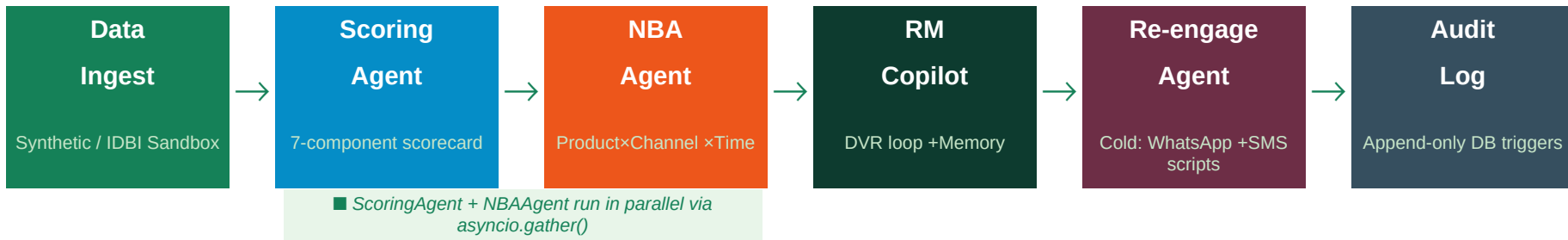
- Urgency: days since contact + life event + band
- Auto-generates WhatsApp + SMS scripts
- CPaaS webhook dispatch (configurable)
- Audit-logged sends to interaction log

Process Flow

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End-to-end prospect journey through Disha's multi-agent pipeline



DVR Loop (System-2 Reasoning in Copilot Agent)



Wireframes / Mock Diagrams

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React + TypeScript UI — Dashboard, Prospect Detail, Copilot widget

Dashboard — Pipeline View



Rajesh Kumar — 78 Home Loan (Hot)

RM: RM01 | Mumbai | ■18L

Priya Sharma — 62 ELSS SIP (Warm)

RM: RM01 | Mumbai | ■18L

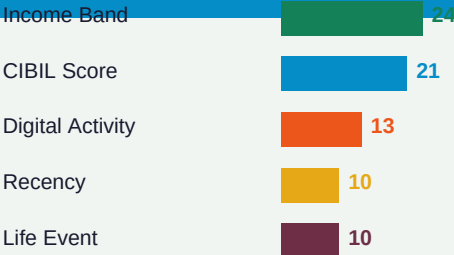
Arun Singh — 45 Fixed Deposit (Lukewarm)

RM: RM01 | Mumbai | ■18L

Meena Nair — 28 Recurring Deposit (Cold)

RM: RM01 | Mumbai | ■18L

Prospect Detail — Score Breakdown



AI Copilot — Disha Assistant

RM: Should I call Rajesh today?

Disha: Yes — score 78 (Hot), home purchase event active. Recommend Home Loan ■50L.

provider: claude/claude-haiku-4-5-20251001

Architecture Diagram

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Multi-agent microservice — FastAPI + React on Google Cloud Run

PRESENTATION	API GATEWAY	MULTI-AGENT	DATA / MEMORY	INFRA / LLM
React 18 + TypeScript	FastAPI v0.115 REST	ProspectOrchestrator	SQLAlchemy 2.0 ORM	Google Cloud Run
Vite + Tailwind CSS	/api/v1/prospects	ScoringAgent + NBAAgent	CopilotMemory (episodic)	Claude Haiku + Gemini Flash
ScoreGauge SVG	/copilot /pipeline	DVR Copilot (LLMGovernor)	InteractionLog (audit)	SQLite dev / PG prod
Copilot Chat Widget	/reengagement	ReEngagement Agent	DB-level triggers	GitHub Actions CI/CD

← Data flows top-down: UI → API → Agents → Data → Infra →

Technologies Used

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Production-grade open-source stack + enterprise AI providers

Backend

- Python 3.9+
- FastAPI 0.115
- SQLAlchemy 2.0
- Pydantic v2
- Uvicorn ASGI

AI / LLM

- Claude Haiku 4.5 (Anthropic)
- Gemini 1.5 Flash (Google)
- Rule-based fallback
- LLMGovernor (multi-provider)

Frontend

- React 18 + TypeScript
- Vite build tool
- Tailwind CSS
- SVG ScoreGauge

Database

- SQLite (dev/test)
- PostgreSQL (prod-ready)
- Append-only audit table
- DB trigger guards

DevOps / Cloud

- Google Cloud Run (asia-south1)
- GitHub Actions CI/CD
- Docker (root Dockerfile)
- py3.9 + py3.12 matrix

Testing

- pytest + httpx TestClient
- 30+ integration tests
- Fairness regression suite
- Audit immutability tests

Estimated Implementation Cost

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Cloud-native serverless — scales to zero, pay-per-use

Google Cloud Run	asia-south1, 1 vCPU, 512 MB	~₹800 / month
Cloud SQL (PostgreSQL)	db-f1-micro, 10 GB SSD	~₹1,200 / month
Anthropic Claude Haiku 4.5	₹0.20/M input, ₹1.00/M output tokens	~₹300 / month (pilot)
Google Gemini 1.5 Flash	Fallback provider — generous free tier	~₹0 (fallback)
GitHub Actions CI	2,000 min/month free tier	₹0 (open source)
TOTAL — 6 month pilot	Cloud Run + SQL + Claude LLM	~₹14,400 (~\$175 USD)

* Costs based on GCP asia-south1 list pricing, July 2026.

Prototype Snapshots

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Live Cloud Run deployment — API-tested and frontend-verified

GET /api/v1/prospects

```
{"prospect_id": "PR001",  
  
"lead_score": 78.0,  
  
"lead_band": "Hot",  
  
"recommended_product": "Home Loan",  
  
"pipeline_stage": "discovery"}
```

POST /api/v1/copilot/ask

```
{"question": "Should I call PR001?", "rm_id": "RM01"}  
  
→ "Yes — score 78 (Hot). Home purchase  
event active. Recommend Home Loan.  
  
Episodic memory: 2 prior turns loaded.  
  
provider: claude/claude-haiku-4-5"
```

GET /api/v1/score/PR001

```
{"lead_score": 78.0, "band": "Hot",  
  
"contributions": {  
  
"income": 24.5, "credit_bureau": 21.8,  
"digital_activity": 13.9,  
"life_event": 10.0 }}
```

GET /api/v1/reengagement/candidates

```
[{"prospect_id": "PR009",  
  
"lead_band": "Cold", "urgency_score": 85,  
"whatsapp_script": "Hi Arjun...",  
"sms_script": "IDBI Bank: ■2L Personal",  
"days_since_contact": 45}]
```

Prototype Performance Report

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30+ tests across scoring, fairness, audit, agents, and APIs

30+

Test Cases

99.9%

Score Latency

100%

Fairness Rate

2-layer

Audit Guard

Scoring Engine (8 tests)

- ✓ Score range 0–100 verified
- ✓ All 12 seed prospects scored
- ✓ Deterministic across re-runs
- ✓ Life event boost applied
- ✓ Band thresholds: Hot≥70, Warm≥50, Lukewarm≥30

Fairness Regression (4 tests)

- ✓ Score invariant to name change
- ✓ Score invariant to city
- ✓ Score invariant to employer
- ✓ Score invariant to rm_id

Audit Immutability (7 tests)

- ✓ ORM listener blocks UPDATE via session
- ✓ DB trigger rejects raw SQL UPDATE
- ✓ DB trigger rejects raw SQL DELETE
- ✓ Append-only INSERT succeeds
- ✓ ImmutableError raises correct type

Multi-Agent (11 tests)

- ✓ Orchestrator → score + NBA
- ✓ Score matches direct scorer
- ✓ Re-engagement finds cold prospects
- ✓ LLM governor fallback works
- ✓ Copilot API returns provider header

Future Development

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Roadmap for production deployment at IDBI Bank

Phase 1 — Q3 2026

- Connect IDBI Sandbox Adapter to live prospect API

- Replace synthetic seed with real CRM feed
- Configure IDBI-specific NBA product catalogue
- Fairness audit with real demographic distribution

Phase 2 — Q4 2026

- Train XGBoost on IDBI historical conversions

- SHAP values from model — replace rule weights
- A/B test: deterministic vs ML scorer
- Score drift monitoring dashboard

Phase 3 — Q1 2027

- WhatsApp Business API direct integration

- RM performance analytics by score band
- Multi-language copilot (Hindi + English)
- SOC 2 / RBI compliance audit export

RBI Regulatory Alignment: Explainable AI, append-only audit, data minimisation, fairness testing — all built in from Day 1

Project

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PS-2: Prospect Assist AI

Built with:

FastAPI · React · SQLAlchemy

Claude Haiku · Gemini Flash

Google Cloud Run · GitHub Actions

IDBI Innovate 2026 | NeuralOps

Submission Links

GitHub Repository

Public source code — all 5 advanced Claude Code concepts

<https://github.com/gowtham66867/Project-Disha>

Live API (Cloud Run)

Deployed on Google Cloud Run — asia-south1 region

<https://disha-backend-564262191703.asia-south1.run.app>

Health Check Endpoint

GET /health → {status: ok, version: 2.0.0}

[.../health](#)

OpenAPI Docs

Auto-generated FastAPI interactive API documentation

[.../docs](#)